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Assignment 5

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 90 = 3 + 87 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

<https://docs.google.com/document/d/1dzAggy7UhyrmR9CylVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing>

I. The Multiplication Rule (8 x 3 = 24 pts)

1) A simple random sample of adults in a metropolitan area was selected and a survey was administered to determine political views. The results are recorded below:

Political Views				
Age	Conservative	Moderate	Liberal	Total
18-29	10	15	30	55
30-44	20	30	35	85
45-59	35	15	20	70
Over 60	20	15	10	45
Total	85	75	95	255

a) What is the probability that a person chosen at random from this sample will have moderate political views?

Answer:

$$75/255=0.294$$

b) What is the probability that a person chosen at random from those in the sample who are between the ages of 30 to 44 will have moderate political views?

Answer:

$$30/85=0.353$$

c) Based on your answers to (a) and (b), are political views and age independent for the population of adults in this metropolitan area? Why or why not?

Answer:

They are not independent because $P(A) \neq P(A|B)$.

2) Given that $P(A) = 0.6$, $P(B) = 0.3$, and $P(B|A) = 0.5$

a) Find $P(A \text{ and } B)$

Answer:

$$0.6 * 0.5 = 0.3$$

b) Find $P(A \text{ or } B)$

Answer:

$$0.6 + 0.3 - 0.5 = 0.4$$

c) Are events A and B independent?

Answer:

A and B are not independent because $P(B) \neq P(B|A)$.

3) At Portillo's in Rockford, the distribution of ages and what shift they work on a Saturday is below. Fill in the table with the missing information.

Age	Open - 1pm	1pm - 7pm	7pm - Close	Total
Under 18	6	2	21	29
18 to 49	4	5	2	11
40 and over	10	5	25	40
Total	20	12	48	80

Are the events "Open - 1pm" and "40 and over" independent? Show your work.

Answer:

A = Open - 1pm

B = 40 and over

$$P(A) = 20/80 = 0.25$$

$$P(A|B) = 10/40 = 0.25$$

$P(A) = P(A|B)$, therefore they are independent.

II. Discrete and Continuous Random Variables (39 pts)

1) A random sample of adults was taken and asked the question “Last week, how many days did you sit down and eat at a restaurant?” X represents the number of days they responded with. Here is the resulting probability model:

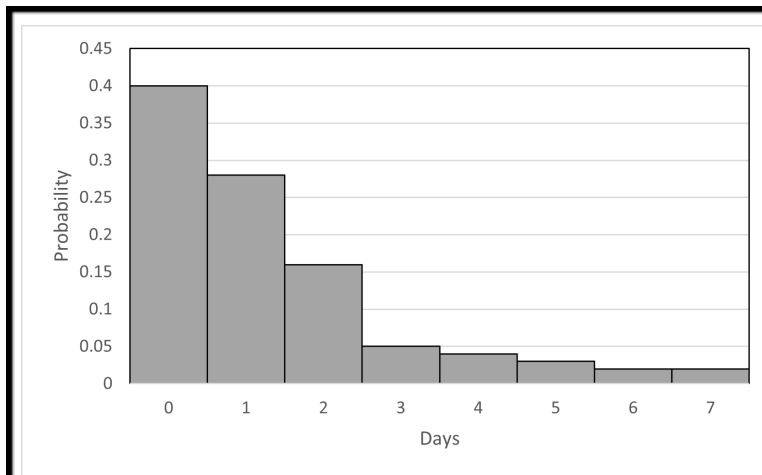
Days (X)	0	1	2	3	4	5	6	7
Probability	0.40	0.28	0.16	0.05	0.04	0.03	0.02	0.02

(a) Show that the probabilities in this distribution add up to 1.

Answer:

$$0.40 + 0.28 + 0.16 + 0.05 + 0.04 + 0.03 + 0.02 + 0.02 = 1$$

(b) Make a histogram of the probability distribution. Describe the shape of the distribution.



The distribution is unimodal right skewed with a center near 0-1 days. There appear to be several outliers of larger value.

(c) Find and interpret $P(X < 4)$.

Answer:

$P(X < 4) = 0.89$. This means a random adult ate at a restaurant less than 4 days last week.

(d) Express the event “Ate out at least twice last week” in terms of X. What is the probability of this event?

Answer:

$$P(X \geq 2) = 0.32$$

(e) Find and interpret the mean.

Answer:

The mean is 1.32 days, which is the average or expected number of days a random adult ate out last week.

(f) Find the standard deviation.

Answer:

The standard deviation is 1.649.

2) A Denver Ski Resort compiles their profit data yearly and has discovered that if it snows less than 60 inches in a season, they will lose \$90,000. If it snows between 60 inches and 80 inches, they will make \$70,000, and if it snows above 80 inches, they will make \$200,000. Let X represent the profit (any losses would be represented as a negative value). If the probability of less than 60 inches is 40%, the probability of between 60 inches and 80 inches is 45%, create a table to represent the probability distribution. Then comment on if this is a profitable business model or not based on the mean of the distribution. [6 pts]

Answer:

Snowfall (inches)	Profit (\$)	Probability	The mean of this distribution is \$25,500 so the business is profitable on average.
<60	-90000	0.4	
60-80	70000	0.45	
>80	200000	0.15	

3) A new type of gambling game uses two dice and the sum of the two dice determines your winnings. It costs \$5 to play the game. If the sum is a 2 or a 12, the player will win \$50. If the sum is a 7, the player will win \$10. Any other sum and the player will lose and not get anything. If X represents the winnings (with -5 representing a loss, since a player will lose that amount of money), create a table to represent the probability distribution. Using the mean of this distribution, do you believe the game is in favor of the player or the "house"? [6 pts]

Answer:

Dice Sum	2	3	4	5	6	7	8	9	10	11	12
Winnings	45	-5	-5	-5	-5	5	-5	-5	-5	-5	45
Probability	0.028	0.056	0.083	0.111	0.139	0.167	0.139	0.111	0.083	0.056	0.028

The mean of the distribution is -0.555 so the game is in favor of the house.

*This assumes the cost to play the game isn't accounted for in the \$50 and \$10 wins, since the question doesn't make it clear. If the player wins \$50 and \$10 plus they get their \$5 back then the mean is 0.555 and the game is in favor of the player instead of the house.

4) Choose an American household at random and let the random variable X be the number of persons living in the household. The probability of X is:

X	1	2	3	4	5	6	7 or more
P(X)	0.25	0.32	0.17	0.15	0.07	0.03	0.01

Find and interpret the following probabilities:

a) $P(X > 3)$

Answer:

$$P(X > 3) = 0.26$$

A random household has a probability of 0.26 that there are more than 3 persons living there.

b) $P(X \leq 4)$

Answer:

$$P(X \leq 4) = 0.89$$

A random household has a probability of 0.89 that there are 4 or fewer persons living there.

c) $P(2 \leq X \leq 5)$

Answer:

$$P(2 \leq X \leq 5) = 0.71$$

A random household has a probability of 0.71 that there are 2 to 5 persons living there.

III. Combining Random Variables (8 x 3 = 24 pts)

1) In the 2021 baseball season, the Brewers runs per game, B, varied from game to game but had $\mu_B = 3.7$ and $\sigma_B = 1.1$. The Cubs runs per game, C, were $\mu_C = 4.9$ and $\sigma_C = 1.6$.

Find the following (show your work):

μ_{B+C} $\mu_{B+C} = \mu_B + \mu_C = 3.7 + 4.9 = 8.6$	μ_{B-C} $\mu_{B-C} = \mu_B - \mu_C = 3.7 - 4.9 = -1.2$
σ_{B+C} $\sigma_{B+C} = \sqrt{(\sigma_B^2 + \sigma_C^2)} = \sqrt{(1.21 + 2.56)} = 1.94$	σ_{B-C} $\sigma_{B-C} = \sqrt{(\sigma_B^2 + \sigma_C^2)} = \sqrt{(1.21 + 2.56)} = 1.94$

2) At our town's ice cream shop, if you order a cone, you can determine the number of scoops you would like. Let X represent the number of scoops ordered and the probability that a random customer would ask for that number of scoops.

X	1	2	3	4	5
P(X)	0.20	0.38	0.28	0.11	0.03

It costs \$2.50 for the cone and \$0.75 for each scoop of ice cream. Find the amount of money the store expects to make on a randomly selected customer. **[6 pts]**

Answer:

$$E(X) = 1 * 0.20 + 2 * 0.38 + 3 * 0.28 + 4 * 0.11 + 5 * 0.03 = 0.20 + 0.76 + 0.84 + 0.44 + 0.15 = 2.39$$
$$2.39 * 0.75 = 1.79$$

The store should expect to make \$4.29 on a randomly selected customer, \$2.50 for the cone and \$1.79 for the scoops of ice cream.

3) Suppose the owners of the ice cream shop want to keep track of their profits. The cost of a cone is \$0.10 for the shop and the cost of a scoop of ice cream is \$0.45. What is the expected profit on a randomly selected customer? Use the information from problem #2 to help you answer this question. Show your work. **[6pts]**

Answer:

The profit for a cone is $\$2.50 - \$0.10 = \$2.40$. The profit for a scoop is $\$0.75 - \$0.45 = \$0.30$. A randomly selected customer purchases 1 cone and 2.39 scoops. Therefore, the profit on a randomly selected customer is $1 * \$2.40 + 2.39 * \$0.30 = \$3.11$.

THE END